

Ahmed Layouni

Final-Year Computer Science Engineering Student | AI & Data Engineering

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Profile

Final-year Computer Science Engineering student at ENSI (Tunisia), specializing in Artificial Intelligence and Data Engineering. Hands-on experience shipping production-grade computer vision, agentic RAG/LLM and full-stack data systems through internships.

Professional Experience

Computer Vision & AI Engineering Intern

WiseVision – Tunis, Tunisia, May – August 2026

- Built a real-time CCTV analytics pipeline (YOLO11m-pose + BoT-SORT/OSNet ReID) counting customer entries, staff–customer interactions and per-zone occupancy.
- Reached **87% accuracy** on entry counting and **90% accuracy** on interaction counting vs. hand-labelled ground truth.
- Deployed on-premise on an **NVIDIA Jetson Orin NX** (TensorRT FP16), running at **3× real-time**.
- **Tech Stack:** YOLO11m-pose, BoT-SORT, OSNet ReID, TensorRT, ONNX, NVIDIA Jetson Orin NX

Full-Stack Software Engineering Intern

MenaML – Remote, Qatar June – August 2026

- Built a 5-portal conference-management platform in a TypeScript/Next.js monorepo (~57K LOC, ~100 pages).
- Designed **40 tRPC routers** exposing ~380 type-safe procedures over a **69-table** PostgreSQL/Drizzle schema.
- Shipped cross-portal features: multi-step travel/visa approvals, RBAC, and dual-backend file storage.
- Ensured reliability with a **526-case** unit/integration/E2E test suite.
- **Tech Stack:** TypeScript, Next.js, tRPC, PostgreSQL, Drizzle ORM, docker, git

ML-Engineering Inter

Orange Tunisia – Tunis, Tunisie, June – July 2025

- Built a one-month weather forecasting model per Tunisian governorate using Temporal Fusion Transformers, reaching **86% prediction accuracy**.
- Designed a full MLOps pipeline: data versioning, experiment tracking and model deployment via MLflow.
- **Tech Stack:** Temporal Fusion Transformer, MLflow, Pytorch, TensorFlow, docker, git

Projects

Agentic RAG System for Technical Documentation

- Built hybrid BM25 + vector retrieval (Reciprocal Rank Fusion) driven by a self-correcting LangGraph agent that grades each retrieved chunk's relevance and rewrites the query before generating a grounded answer.
- Engineered a two-stage chunking pipeline (markdown header-aware splitting + recursive character splitting) with contextual breadcrumb tagging across 155 docs → 2,539 chunks; fixed a header-anchor parsing bug that was corrupting breadcrumbs on 123 files.
- **Tech Stack:** LangChain, LangGraph, vector embedding, Pydantic, FastAPI, ChromaDB, BM25, docker, LLM

Smart Glasses for the Visually Impaired

- Architected and deployed a cloud pipeline combining object detection (YOLOv8), OCR (Mistral) and currency recognition, controlled by voice commands.
- Built a Flutter app with real-time text-to-speech output; **92% detection accuracy, <1.5s latency**, tested on 500+ real images.
- **Tech Stack:** YOLOv8, Mistral OCR, Cloud API, Flutter, TTS, STT

Data Engineering – Real-Time Pipeline

- Built a scalable real-time data pipeline (Apache Spark, Databricks) ingesting and transforming 1M+ events from diverse sources (APIs, databases, message queues), with fault-tolerant architecture, schema validation and quality checks via Delta Lake.
- Implemented incremental processing with Spark Structured Streaming to reduce latency and optimize compute costs, and deployed automated end-to-end orchestration workflows (Databricks Workflows, Delta Live Tables) for production-grade reliability.
- **Tech Stack:** Apache Spark, Databricks, Delta Lake, Python/PySpark, SQL, Structured Streaming, Airflow/DLT, Pandas, dbt, Spark SQL

Financial Analysis & Stock Price Prediction

- Built a financial agent predicting the future value of any global stock and automatically constructing a personalized investment portfolio.
- **Tech Stack:** LangChain, OpenAI, Thesys, React, Node.js

Education

ENSI – National School of Computer Science – Engineering Degree in Computer Science (In progress)	2024 – 2027
IPEIS – Preparatory Classes, Mathematics & Physics (Rank: 324th/1750)	2022 – 2024
Sfax Pilot High School – Baccalaureate in Mathematics, Very Good Mention (17.32/20)	2018 – 2022

Technical Skills

Programming Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL

AI / ML / Data: YOLOv8/YOLO11, LightGBM, Temporal Fusion Transformer, LangChain, LangGraph, OpenAI API, Mistral OCR, Google Gemini AI, ChromaDB, BM25/Hybrid Retrieval, BoT-SORT, OSNet Re-ID, TensorRT, NVIDIA Jetson Orin NX, Apache Spark, Databricks

Web & Backend: Next.js, tRPC, React, Node.js, Drizzle ORM